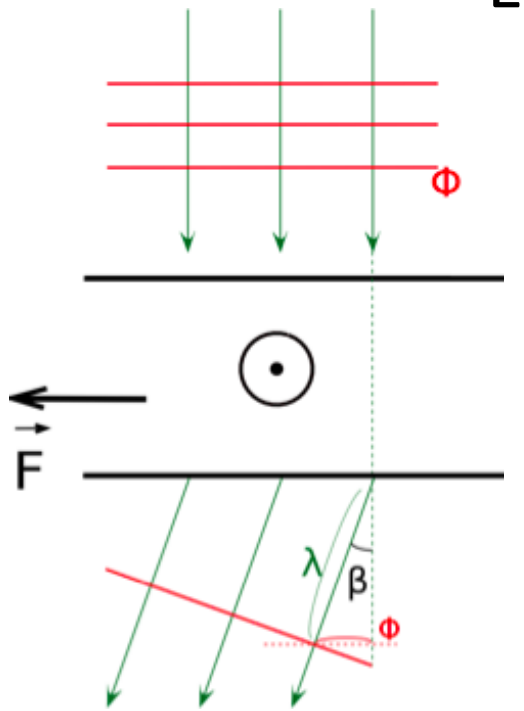
Electron Holography



b ~ qq mrad

Wave approach to Aharonov-Bohm

$$\phi = C_E \int_{\infty} V_{int}(\vec{r}, z)dz - \frac{e}{\hbar}\int_{\infty} A_z(\vec{r}, z)dz$$

The electric action of the sample is taken into account through the mean inner potential of the studied material.

The constant CE depends on the electron energy and therefore on the microscope used.

Takeshi Kasama, Rafal E. Dunin-Borkowski and Marco Beleggia (2011). Electron Holography of Magnetic Materials, Holography - Different Fields of Application, Dr. Freddy Monroy (Ed.), ISBN: 978-953-307-635-5, InTech, DOI: 10.5772/22366. Available from: http://www.intechopen.com/books/holography-different-fields-of-application/electron-holography-of-magnetic-materials

Rafal E. Dunin-Borkowski, Takeshi Kasama, Marco Beleggia, Giulio Pozzi, Ch.6: Lorentz Microscopy and Electron Holography of Magnetic Materials, in Handbook of Nanoscopy, Vol.1, Gustaaf Van Tendeloo, Dirk Van Dyck, Stephen J. Pennycook, Eds, Wiley-VCH (2012)
DOI: 10.1002/9783527641864.ch6
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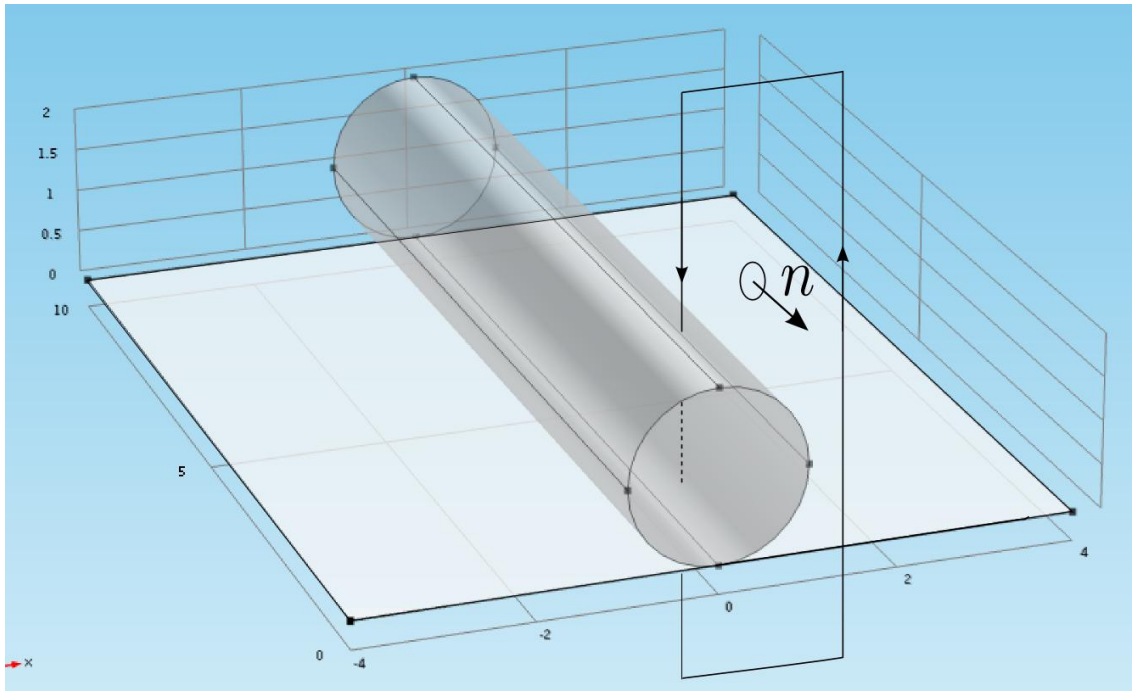
First approach to calculate the magnetic phase



Pure 3D calculations, $$\Gamma = \oint_{\partial D} \mathbf{dl} \cdot \mathbf{A} = \iint_D dS\ \mathbf{curl}\ \mathbf{A} \cdot \mathbf{n} = \iint_D dS\ \mathbf{B} \cdot \mathbf{n}$$

Biziere et al, Nano Lett. 2013, 13, 2053–2057

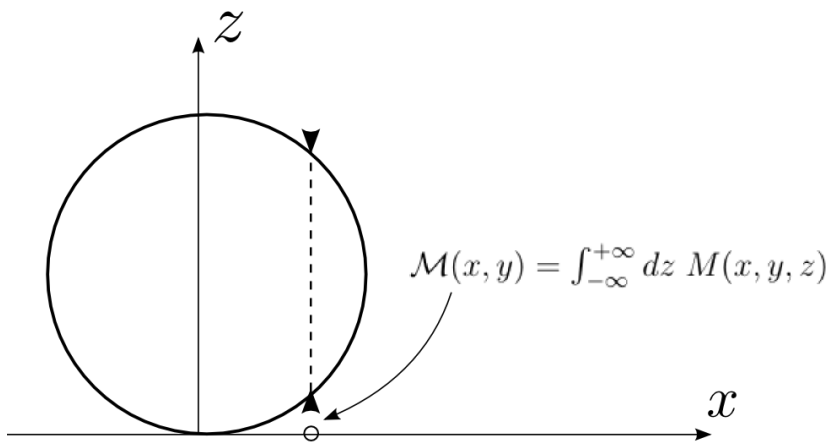
- One must compute the flux through the vertical surface for each point of the horizontal plane. These calculations are highly time-consuming
- The support D must be large enough to limit finite-size effects.

Second approach to calculate the magnetic phase

- Reducing the problem from 3D to 2D yields
a significant reduction in computational cost.

- The size effects are not due to the phase calculation
but rather to the finite size of the micromagnetic system.

- One implements a 2D integral method that requires
computing an integral of the magnetization along the beam path.



It allows us to compute the circulation of $\mathbf{A}$ along a vertical path crossing the (Oxy) plane at $q(x, y)$:

$$\Gamma(q) = \int_{-\infty}^{+\infty} dz_Q \; A_z(Q) = \frac{\mu_0}{4\pi} \int d^3 r_P \; (e_z \times M(P)) \cdot \int_{-\infty}^{+\infty} dz_Q \; \frac{PQ}{|PQ|^3} \tag{1}$$

To simplify subsequent calculations, we define the normalized magnetization $m_\perp(p)$, orthogonal to Oz, as follows :

$$m_\perp(p) = \frac{1}{M_\mathrm{s} \; t(p)} \; e_z \times \int_{-\infty}^{+\infty} dz \; M(P) = \frac{1}{M_\mathrm{s} \; t(p)} \; e_z \times \mathcal{M}(x, y) \tag{2}$$

where $t(p)$ is the thickness of the magnetic material at the point $p(x, y)$, the orthogonal projection of P onto the (Oxy) plane.

After applying Gauss's theorem to the last integral, one obtains :

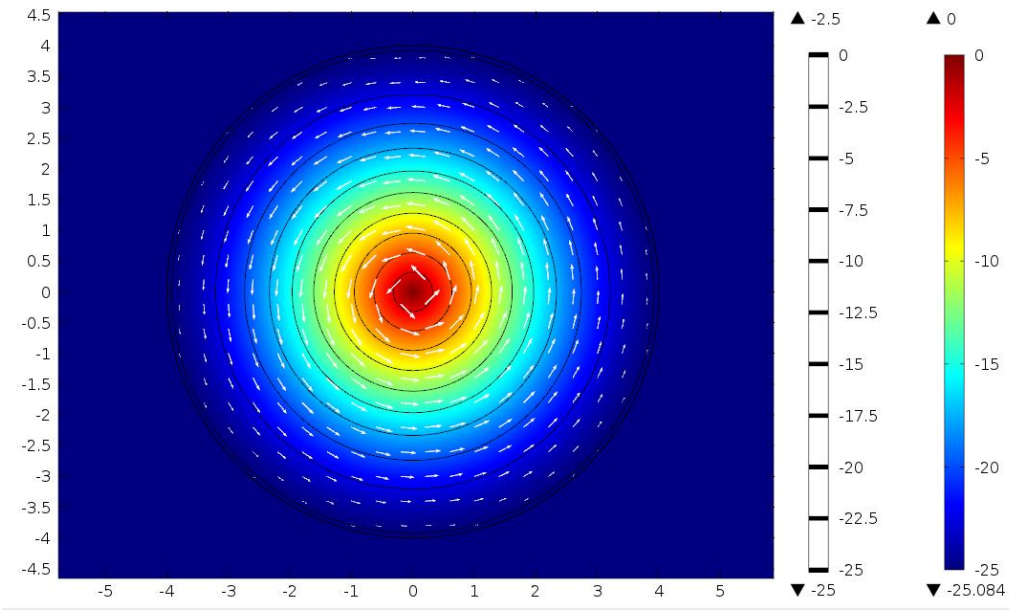
$$\Gamma(q) = -\frac{\mu_0 M_s}{2\pi} \int\limits_{\mathbb{R}^2} dx_p \; dy_p \; t(p) \; m_\perp(p) \cdot \frac{pq}{|pq|^2} \tag{3}$$

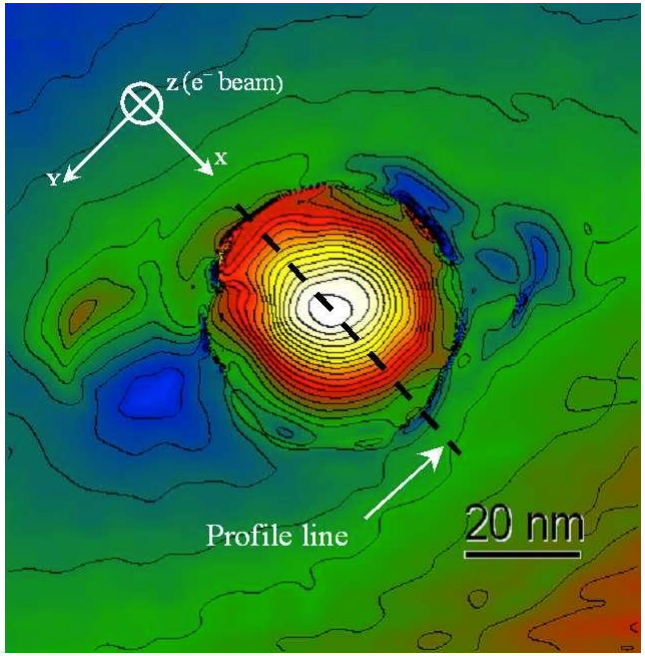
In electron holography, the magnetic phase is defined by $\phi_m = -\frac{e}{\hbar}\Gamma$

Cas test : sphère à aimantation de type curling dans le plan

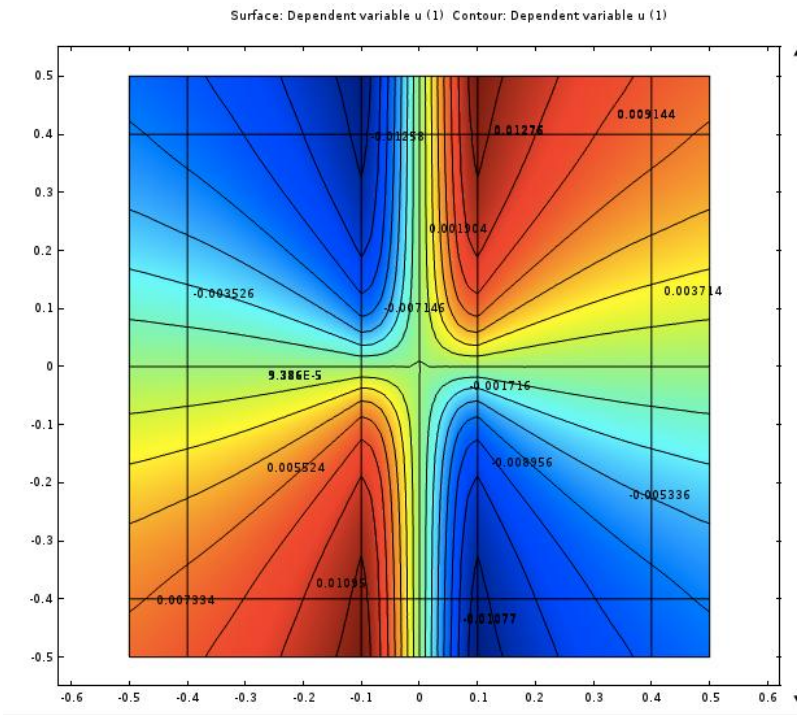
$$\Gamma(x,y) = -\mu_0 M R^2 \left(\arcsin\left(\frac{h(x,y)}{R}\right) + \frac{h(x,y)}{R}\sqrt{1-\left(\frac{h(x,y)}{R}\right)^2} \right)$$

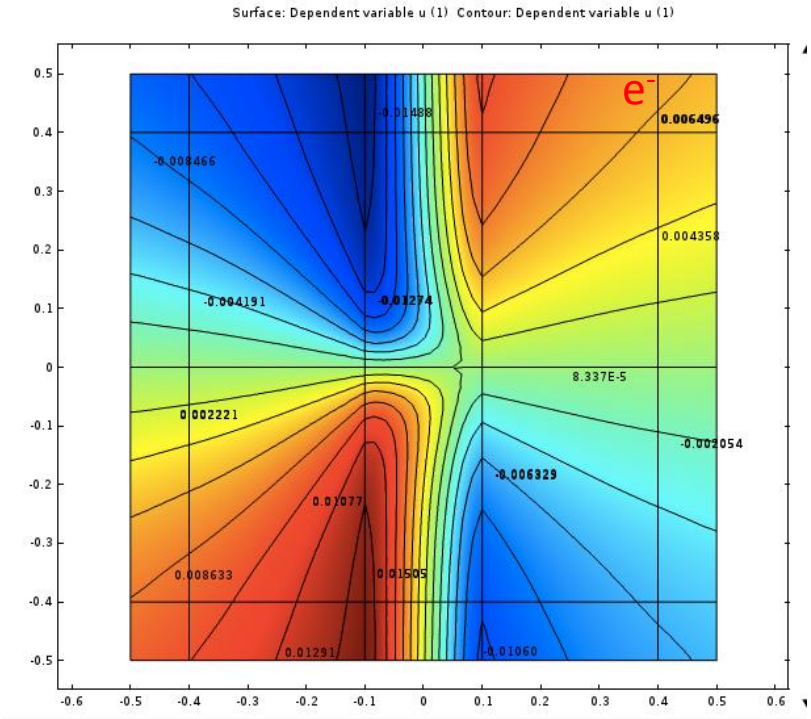
avec $h(x,y) = \min(\sqrt{x^2+y^2},\ R)$.



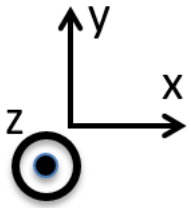


With an ansatz for the TW magnetization distribution





$$\begin{cases} m_x = 0 \\ m_y = \tanh(y/\delta) \\ m_z = 1/\cosh(y/\delta) \end{cases}$$



$$\begin{cases} m_x = 1/\cosh(y/\delta) \\ m_y = \tanh(y/\delta) \\ m_z = 0 \end{cases}$$